unfair. What is needed is the scientific and reliable information about the source of pollutants on a continuing basis and the technologies that will work to reduce pollution from them.

21. Which one of the following statements best reflects the most logical and rational implication conveyed by the passage?

(a) Arbitrary curbs on vehicles to reduce pollution are difficult to implement.
(b) Knee-jerk reactions cannot solve the problem of pollution but an evidence-based approach will be more effective.
(c) A heavy penalty should be enforced on those driving without periodic pollution tests.
(d) In the absence of laws to deal with the problems of pollution, the administration tends to make arbitrary decisions.

Ans. (b) : According to the passage, knee-jerk reactions cannot solve the problem of pollution but an evidence based approach will be more effective.

Passage-2

Good corporate governance structures encourage companies to provide accountability and control. A fundamental reason why corporate governance has moved not the economic and political agenda worldwide has been the rapid growth in international capital markets. Effective corporate governance enhances access to external financing by firms, leading to greater investments, higher growth and employment, Investors look to place their funds where the standards of disclosure, of timely and accurate financial reporting, and of equal treatment to all stakeholders are met.

22. Which of the following statements best reflects the logical inference from the passage given above?

(a) It is an important agenda of the countries around the world to ensure access to good external financing.
(b) Good corporate markets ensure that the firms maintain good corporate governance.
(c) International capital markets ensure that the firms maintain good corporate governance.
(d) Good corporate governance paves the way for robust supply chains.

Ans. (b) : The statement best reflects the logical inference from the passage is - "Good corporate governance improves the credibility of the firm".

Passage-3

Elephants are landscape architects, creating clearings in the forest, preventing overgrowth of certain plant species and allowing space for the regeneration of others, which in turn provide sustenance to other herbivorous animals. Elephants eat plants, fruits and seeds, propagating the seeds when they defecate in other places as they travel. Elephant dung provides nourishment to plants and animals and acts as a breeding ground for insects. In times of drought, they access water by digging holes which benefits other wildlife.

23. Which one of the following statements best reflects the most logical and rational inference that can be drawn from the passage?

(a) The home range of elephants needs to be a vast area or rich biodiversity.
(b) Elephants are the keystone species and they benefit the biodiversity.
(c) Rich biodiversity cannot be maintained in the forests without the presence of elephants.
(d) Elephants are capable of regenerating forests with species as per their requirement.

Ans. (b) : The statement best reflects the most logical and rational inference that can be drawn from the passage is -
"Elephants are the keystone species and they benefit the biodiversity".

24. If 7 ⊕ 9 ⊕ 10 = 8, 9 ⊕ 11 ⊕ 30 = 5, 11 ⊕ 17 ⊕ 21 = 13 what is the value of 23 ⊕ 4 ⊕ 15?

(a) 6 (b) 8
(c) 13 (d) 15

Ans. (a) : According to the given,
(i) 7 ⊕ 9 ⊕ 10
It can be written as,
7 + 9 + 10 = 26
and now, 2 + 6 = 8
(ii) 9 ⊕ 11 ⊕ 30
It can be written as,
9 + 11 + 30 = 50
Now, 5 + 0 = 5
(iii) 11 ⊕ 17 ⊕ 21
It can be written as,
11 + 17 + 21 = 49
Now, 4 + 9 = 13
Therefore, 23 ⊕ 4 ⊕ 15
⇒ 23 + 4 + 15 = 42
Now, 4 + 2 = 6

25. Let x be a positive integer such that 7x + 96 is divisible by x. How many values of x are possible?

(a) 10 (b) 11
(c) 12 (d) Infinitely many

Ans. (c) : According to the given,
It 7x+96 is to be divisible by x, 96 should be divisible by x
$96 = 2^5 \times 3$
So number of factors of 96 = (5+1) (1+1)
= 12
Hence, there are 12 possible values of x.

26. If p, q, r and s are distinct single digit positive numbers then what is the greatest value of (p + q) (r + s)?

(a) 230 (b) 225
(c) 224 (d) 221

Ans. (b) : According to the given,
Possible values of p, q, r and s can be 9, 8, 7 and 6 (not in the same order)
Then, p + q + r + s = 30
Product of two numbers is highest, is when both the numbers are equal
9 + 6 = 15
and 8 + 7 = 15
Therefore, 15 × 15 = 225
225 is the greatest value of (p+q) (r+s).

27. A number N is formed by writing 9 for 99 times. What is the remainder if N is divided by 13?

(a) 11 (b) 9
(c) 7 (d) 1

Ans. (a) : According to question, $\frac{9999.....}{13}$

$\frac{9}{13} \Rightarrow Q = 0, \quad R = 9$

$\frac{99}{13} \Rightarrow Q = 7, \quad R = 8$

$\frac{999}{13} \Rightarrow Q = 76, \quad R = 11$

$\frac{9999}{13} \Rightarrow Q = 769, \quad R = 2$

$\frac{99999}{13} \Rightarrow Q = 7692, \quad R = 3$

$\frac{999999}{13} \Rightarrow Q = 76923, \quad R = 0$

$\frac{9999999}{13} \Rightarrow Q = 769230, \quad R = 9$

Similarly, after every 6 divisions, the sequence, of the remainder will be repeated
Therefore, 16 × 16 = 96
96 ⇒ R = 0
97 ⇒ R = 9
98 ⇒ R = 8
99 ⇒ R = 11
Hence, remainder will be 11.

28. Each digit of a 9-digit number is 1. It is multiplied by itself. What is the sum of the digits of the resulting number?

(a) 100 (b) 80
(c) 81 (d) 100

Ans. (c) : As we know,
$(11)^2 = 121$
$(111)^2 = 12321$ and so an
Therefore, $(111111111)^2 = 1234567898765321$
Sum of digits = 81

29. What is the sum of all digits which appear in all the integers from 10 to 100?

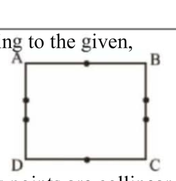
(a) 855 (b) 856
(c) 910 (d) 911

Ans. (b) : According to the given,
Firstly, consider integers from 10 to 99
Sum of all the digits at unit's place = (1+2+....+9) × 9
= 45 × 9
= 405
Sum of all the digits at ten's place = 10(1+2+....+9)
= 450
Hundred's place = 1
Therefore, sum of all the digits = 405 + 450 + 1
= 856

30. ABCD is a square. One point on each of AB and CD; and two distinct points on each of BC and DA are chosen. How many distinct triangles can be drawn using any three points as vertices out of these six points?

(a) 16 (b) 18
(c) 20 (d) 24

Ans. (c) : According to the given,



No three of the six points are collinear.
Then, total number of triangles = $^6C_3$
$= \frac{6!}{3!\ 3!} = 20$

Passage-1

In India, the segregation of municipal waste at source is rare. Recycling is mostly with the informal sector. More than three-fourths of the municipal budget goes into collection and transportation, which leaves very little for processing/resource recovery and disposal. Where does waste-to-energy fit into all this? Ideally it fits in the chain after segregation (between wet waste and the rest), collection, recycling, and before getting to the landfill. Which technology is most appropriate in converting waste (that is biodegradable versus non-biodegradable component) and its calorific value. The biodegradable component of India's municipal solid waste is a little over 50 per cent, and biomethanation offers a major solution for processing this.

31. Based on the above passage, the following assumptions have been made:

1. Collection, processing and segregation of municipal waste should be with government agencies.
2. Resource recovery and recycling require technological inputs that can be best handled by private sector enterprises.

Which of the assumptions given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans. (d) : Neither statement 1 nor 2 given above are correct.

32. Which one of the following statements best reflects the crux of the passage?

(a) Generation of energy from municipal solid waste is inexpensive.
(b) Biomethanation is the most ideal way of generating energy from municipal solid waste.
(c) Segregation of municipal solid waste is the first step in ensuring the success of water-to-energy plants.
(d) The biodegradable component of India's municipal solid waste is not adequate to provide energy from waste efficiently/ effectively.

Ans. (c) : Statement which best reflects the crux of passage is -
"Segregation of municipal solid waste is the first step in ensuring the success of waste to energy plants".

Passage-2

There is a claim that organic farming is inherently safer and healthier. The reality is that because the organic farming industry is still young and not well-regulated in India, farmers and consumers, alike, are not only confused about what products are best for them, but